

Physics World Models for Computational Imaging: A Universal Physics-Information Law for Recoverability, Carrier Noise, and Operator Mismatch

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Abstract

Computational imaging systems routinely fail in practice because the assumed forward model diverges from the true physics, yet no existing framework systematically diagnoses *why* reconstruction degrades. We introduce Physics World Models (PWM), a universal diagnostic and correction framework grounded in the TRIAD LAW: every imaging failure decomposes into exactly three root causes—recoverability loss (**Gate 1**), carrier-noise budget violation (**Gate 2**), and operator mismatch (**Gate 3**). PWM compiles 64 modalities spanning five physical carriers (photons, electrons, spins, acoustic waves, and particles) into a unified OPERATORGRAPH intermediate representation comprising 89 validated operator templates. Autonomous, deterministic agents diagnose the dominant failure gate and correct the forward model without retraining any reconstruction algorithm. Across 9 fully validated modalities (16 registered), correction yields improvements ranging from +3.55 dB to +48.25 dB. **Gate 3** is identified as the dominant bottleneck in every validated modality, demonstrating that a decade of solver-centric progress has overlooked the principal source of imaging failure. The TRIAD LAW provides the first universal, quantitative language for imaging diagnosis.

Introduction

Why do state-of-the-art reconstruction algorithms fail in practice? The answer is deceptively simple: the assumed forward model is wrong, and nobody measures this systematically. The computational imaging community has devoted extraordinary effort to designing ever more powerful solvers—from compressed sensing^{1,2} and plug-and-play priors³ to end-to-end deep unrolling networks⁴—while treating the forward model as a fixed, trusted input. This implicit assumption is rarely justified. Optical masks shift during assembly, MRI coil sensitivities drift with patient positioning, and CT geometries deviate from their nominal calibration. When these mismatches arise, even the most sophisticated reconstruction algorithms collapse, and the resulting artifacts are routinely misattributed to solver limitations rather than to their true cause: an incorrect physics model.

32 The scale of this crisis is striking. Consider coded aperture snapshot spectral imaging
 33 (CASSI), a representative photon-domain modality. Under ideal conditions—where the
 34 true coded mask is known exactly—the state-of-the-art transformer solver MST-L⁵ achieves
 35 34.81 dB on a standard benchmark⁶. Introduce a realistic perturbation of just one pixel of
 36 lateral mask shift, and MST-L plummets to 18.09 dB, a catastrophic loss of 16.72 dB. To
 37 put this in perspective, the cumulative improvement from a decade of solver development
 38 in CASSI—progressing from early iterative methods through deep unrolling to modern
 39 transformer architectures—amounts to roughly 2–3 dB. A single pixel of undiagnosed mask
 40 shift erases more than five times the gains of an entire research generation. This is not
 41 a pathological edge case; analogous degradations appear across modalities, from lensless
 42 imaging to magnetic resonance imaging^{7,8} to computed tomography⁹.

43 The root problem is a missing diagnostic layer. When a reconstruction fails, the prac-
 44 titioner faces a differential diagnosis with at least three distinct failure modes. First, the
 45 measurement may be fundamentally information-deficient: the null space of the forward
 46 operator may preclude recovery regardless of the solver or signal-to-noise ratio. Second,
 47 the carrier budget may be insufficient: too few photons, too low a dose, or too short an
 48 acquisition may push the measurement below the quantum or thermal noise floor. Third,
 49 the assumed forward model may diverge from the true physics: the operator used for recon-
 50 struction may not match the operator that generated the data. These three failure modes
 51 interact, compound, and masquerade as one another, yet no existing framework disentangles
 52 them.

53 Previous work has addressed fragments of this problem. Calibration methods exist for
 54 specific instruments^{10,11}, but they are modality-specific and do not generalise. Uncertainty
 55 quantification techniques can flag unreliable reconstructions, but they do not diagnose the
 56 *cause* of the unreliability. Robustness studies perturb individual systems¹², but they lack a
 57 unifying formalism that connects perturbation types across the electromagnetic, acoustic,
 58 and particle-physics domains. The imaging community thus remains in a pre-diagnostic
 59 era: systems are built, they fail, and the failure is addressed *ad hoc* if it is addressed at all.

60 This paper introduces Physics World Models (PWM), a universal framework that ele-
 61 vates imaging diagnosis to a first-class computational task alongside reconstruction. The
 62 theoretical backbone of PWM is the TRIAD LAW, which asserts that every imaging failure
 63 decomposes into exactly three root causes, termed gates: **Gate 1** (recoverability), **Gate 2**
 64 (carrier budget), and **Gate 3** (operator mismatch). The TRIAD LAW is not a heuristic; it is
 65 a structured decomposition grounded in the information-theoretic and physical constraints
 66 that govern all linear inverse problems. For every modality and every reconstruction fail-
 67 ure, PWM produces a TRIADREPORT: a mandatory diagnostic artifact that identifies the
 68 dominant gate, quantifies the evidence, and prescribes a corrective action.

69 To apply the TRIAD LAW across the full landscape of computational imaging, PWM in-
 70 troduces the OPERATORGRAPH intermediate representation (IR): a directed acyclic graph

(DAG) formalism that compiles forward models from 64 modalities spanning five physical carriers—photons, electrons, spins, acoustic waves, and particles—into a common computational substrate. Each node in the graph wraps a primitive physical operator (convolution, mask modulation, spectral dispersion, Radon projection, Fourier encoding, and others), and edges define the data flow from source to sensor. The OPERATORGRAPH IR currently comprises 89 validated templates, enabling PWM to reason about imaging systems as diverse as coded aperture spectral imaging¹³, ptychographic electron microscopy¹⁴, accelerated MRI¹⁵, photoacoustic tomography, and neutron computed tomography within a single formalism.

Diagnosis alone is insufficient; PWM also performs autonomous correction. Three deterministic agents—**RecoverabilityAgent**, **PhotonAgent**, and **MismatchAgent**—evaluate each gate without requiring any large language model or learned component. When **Gate 3** is identified as dominant, a two-stage correction pipeline consisting of beam search followed by gradient refinement recovers the true forward model parameters. Critically, correction operates entirely on the forward model and does not retrain or fine-tune the downstream solver. Across 9 fully validated modalities (with 7 additional modalities registered for future validation), autonomous correction yields improvements ranging from +3.55 dB to +48.25 dB. In every validated modality, **Gate 3** is identified as the dominant failure gate, confirming that operator mismatch—not solver weakness or noise—is the principal bottleneck in modern computational imaging.

The Triad Law

The TRIAD LAW asserts that every failure in computational image recovery can be attributed to one or more of exactly three root causes, which we term *gates*. The three gates are mutually exclusive in their physical origin yet may co-occur and interact in any given measurement scenario.

Gate 1: Recoverability. **Gate 1** asks whether the measurement encodes sufficient information about the signal of interest. Formally, if the forward operator $H \in \mathbb{R}^{m \times n}$ maps the unknown signal $\mathbf{x} \in \mathbb{R}^n$ to the measurement $\mathbf{y} = H\mathbf{x} + \mathbf{n}$, then the null space $\mathcal{N}(H)$ defines the set of signal components that are fundamentally invisible to the sensor. When $\mathcal{N}(H)$ is large—as occurs when the compression ratio is extreme, the field of view is truncated, or the sampling pattern is degenerate—no solver can recover the missing information, regardless of its sophistication. **Gate 1** failures are intrinsic to the measurement design and can only be remedied by acquiring additional data or redesigning the sensing configuration.

Gate 2: Carrier Budget. **Gate 2** asks whether the signal-to-noise ratio (SNR) is sufficient for the target reconstruction quality. Every physical carrier—photons, electrons, spins,

acoustic waves, particles—is subject to fundamental noise limits: shot noise for photon-counting systems, thermal noise in electronic detectors, T_1/T_2 relaxation noise in magnetic resonance. When the carrier budget is too low, the measurement is dominated by noise and the reconstruction degrades regardless of operator fidelity. **Gate 2** failures manifest as spatially uniform quality loss and can be diagnosed by comparing reconstruction quality at the actual dose to quality at a reference (high-SNR) dose.

Gate 3: Operator Mismatch. **Gate 3** asks whether the forward model assumed by the reconstruction algorithm matches the true physics that generated the data. Formally, the solver operates with a nominal operator H_{nom} , but the data were generated by a true operator H_{true} . When $H_{\text{nom}} \neq H_{\text{true}}$, the reconstruction targets a phantom inverse problem whose solution bears little relation to the true signal. **Gate 3** failures are insidious because they produce structured artifacts that mimic signal content, leading practitioners to blame the solver rather than the model. Sources of mismatch include geometric misalignment (mask shift, rotation, magnification error), parameter drift (coil sensitivity variation, gain instability), and model simplification (ignoring diffraction, neglecting scattering, linearising a nonlinear process).

Mathematical formulation. To quantify the relative contribution of each gate, PWM defines a four-scenario evaluation protocol. Let PSNR_{I} denote reconstruction quality under ideal conditions (true operator, high SNR), PSNR_{II} under mismatch conditions (nominal operator applied to data generated by the true operator), and PSNR_{III} under oracle correction (true operator restored). The recovery ratio is:

$$\rho = \frac{\text{PSNR}_{\text{III}} - \text{PSNR}_{\text{II}}}{\text{PSNR}_{\text{I}} - \text{PSNR}_{\text{II}}}, \quad (1)$$

where $\rho = 1$ indicates that the full degradation is attributable to **Gate 3** and is completely recoverable through operator correction, while $\rho = 0$ indicates that the degradation persists even with a perfect operator, implicating **Gate 1** or **Gate 2**.

TriadReport. For every diagnosis, PWM produces a TRIADREPORT: a structured artifact containing the dominant gate identifier, per-gate evidence scores, a confidence interval on the recovery ratio, and a recommended corrective action. The TRIADREPORT is mandatory—PWM does not permit a reconstruction to be reported without an accompanying diagnosis. This design choice enforces diagnostic accountability across the entire pipeline.

Key finding: Gate 3 dominates. Across the 9 modalities for which we have completed full validation, **Gate 3** is the dominant failure gate in every case. In CASSI, a 1-pixel mask shift degrades MST-L from 34.81 dB to 18.09 dB—a loss of 16.72 dB that far exceeds the

~2 dB improvement achievable by upgrading from an iterative solver to a state-of-the-art transformer. The pattern holds beyond photon-domain modalities. In accelerated MRI, a 5% coil sensitivity mismatch produces degradation comparable to halving the acceleration factor. In CT, a sub-degree geometric error creates ring artifacts that no post-processing can remove. The TRIAD LAW reveals that the imaging community has been optimising the wrong variable: solver improvements yield diminishing returns when the dominant bottleneck is operator fidelity.

OperatorGraph IR

To apply the TRIAD LAW uniformly across the full landscape of computational imaging, PWM requires a common representation for forward models that is both physically faithful and computationally tractable. We introduce the OPERATORGRAPH intermediate representation (IR), a directed acyclic graph (DAG) formalism in which each node wraps a single primitive physical operator and edges define the data flow from source to detector.

Primitive operators. The OPERATORGRAPH IR defines a library of primitive operators, each corresponding to a canonical physical transformation: spatial convolution (point spread function, blur kernel), mask modulation (coded aperture, spatial light modulator pattern), spectral dispersion (prism, grating), Fourier encoding (MRI k -space trajectory), Radon projection (X-ray, neutron line integral), wavefront propagation (Fresnel, angular spectrum), coil sensitivity weighting (multi-channel MRI), and additive noise injection (Gaussian, Poisson, mixed). Every primitive implements both a `forward()` method and an `adjoint()` method, with a validated adjoint consistency check ensuring $\langle H\mathbf{x}, \mathbf{y} \rangle = \langle \mathbf{x}, H^\dagger \mathbf{y} \rangle$ to within numerical precision.

DAG construction. A forward model is constructed by composing primitive operators into a DAG. For example, the CASSI¹³ forward model is represented as Source $\rightarrow$ MaskModulation $\rightarrow$ SpectralDispersion $\rightarrow$ SensorIntegration $\rightarrow$ PoissonNoise. MRI⁷ becomes Source $\rightarrow$ CoilSensitivity $\rightarrow$ FourierEncoding $\rightarrow$ Undersampling $\rightarrow$ GaussianNoise. CT¹⁶ is compiled as Source $\rightarrow$ RadonProjection $\rightarrow$ DetectorResponse $\rightarrow$ PoissonNoise. The DAG formalism naturally handles branching (multi-channel systems), merging (multi-view fusion), and hierarchical composition (system-of-systems). Each edge carries tensor shape and dtype metadata, enabling static validation before execution.

Five physical carriers. The OPERATORGRAPH IR is organised around five physical carrier families: *photons* (visible, infrared, X-ray, gamma), *electrons* (scanning, transmission, diffraction), *spins* (nuclear magnetic resonance, electron spin resonance), *acoustic waves* (ultrasound, photoacoustic), and *particles* (neutrons, protons, muons). Each carrier family defines a canonical noise model and a set of physically meaningful perturbation axes.

174 The carrier abstraction ensures that the TRIAD LAW diagnostic agents operate identically
 175 regardless of the underlying physics.

176 **Physics Fidelity Ladder.** Not all applications require the same level of physical fidelity.
 177 The OPERATORGRAPH IR defines a four-tier Physics Fidelity Ladder: Tier 1 (linear, shift-
 178 invariant approximation), Tier 2 (linear, shift-variant), Tier 3 (nonlinear, ray-based or
 179 wave-based), and Tier 4 (full-wave simulation or Monte Carlo transport). Each tier inherits
 180 the operator interface and adjoint contract from its parent, enabling solvers to operate
 181 transparently across fidelity levels. For the 64 modalities compiled in this work, Tier 1 and
 182 Tier 2 models suffice for diagnostic purposes; Tier 3 and Tier 4 are reserved for high-fidelity
 183 correction refinement.

184 **Scale and validation.** The current OPERATORGRAPH library contains 89 validated tem-
 185 plates spanning 64 distinct imaging modalities. Validation consists of three automated
 186 checks: adjoint consistency ($|\langle H\mathbf{x}, \mathbf{y} \rangle - \langle \mathbf{x}, H^\dagger \mathbf{y} \rangle| < 10^{-6}$), gradient flow (backpropaga-
 187 tion through the full DAG), and dimensional consistency (static shape inference matches
 188 runtime shapes). All 89 templates pass all three checks. The OPERATORGRAPH IR is im-
 189 plemented in Python with a PyTorch backend, enabling seamless integration with existing
 190 deep-learning reconstruction pipelines.

191 Autonomous Diagnosis and Correction

192 PWM performs diagnosis and correction through three specialised agents, each targeting one
 193 gate of the TRIAD LAW. All agents are fully deterministic—they require no large language
 194 model, no learned parameters, and no human intervention.

195 **RecoverabilityAgent (Gate 1).** The **RecoverabilityAgent** evaluates whether the mea-
 196 surement configuration encodes sufficient information. It computes the effective compres-
 197 sion ratio m/n (measurements over unknowns), estimates the null-space dimension via
 198 randomised SVD, and checks for pathological sampling patterns (clustered k -space trajec-
 199 tories, degenerate mask patterns). The output is a recoverability score $s_1 \in [0, 1]$, where
 200 $s_1 < 0.3$ flags a **Gate 1**-dominated failure and triggers a recommendation to increase the
 201 measurement budget.

202 **PhotonAgent (Gate 2).** The **PhotonAgent** evaluates carrier-budget sufficiency. For
 203 photon-domain modalities, it estimates the per-pixel photon count from the measurement
 204 statistics, computes the Cramér–Rao lower bound on reconstruction error, and compares
 205 the achievable SNR to the target quality. For non-photon carriers, analogous estimators
 206 are used: thermal noise variance for MRI, dose-dependent variance for CT, and bandwidth-

207 limited SNR for acoustic modalities. The output is a budget score $s_2 \in [0, 1]$, where $s_2 < 0.3$
 208 indicates a **Gate 2**-dominated failure.

209 **MismatchAgent (Gate 3).** The **MismatchAgent** is the most consequential agent, re-
 210 flecting the empirical dominance of **Gate 3**. It operates in two phases. In the detection
 211 phase, it compares the residual statistics $\|\mathbf{y} - H_{\text{nom}}\hat{\mathbf{x}}\|$ against the expected noise distribu-
 212 tion: systematic residual structure indicates model mismatch. In the localisation phase, it
 213 identifies which operator node in the OPERATORGRAPH DAG is the source of the mismatch
 214 by sweeping perturbations through each node independently and measuring the sensitivity
 215 of the residual. The output is a mismatch score $s_3 \in [0, 1]$ and a pointer to the offending
 216 node.

217 **Correction pipeline.** When **Gate 3** is identified as dominant, PWM activates a two-
 218 stage correction pipeline. **Algorithm 1 (Beam Search)** performs a coarse grid search
 219 over the declared mismatch parameter family $\psi = (\psi_1, \dots, \psi_k)$ associated with the offending
 220 operator node. The parameter family is declared in the OPERATORGRAPH template (*e.g.*,
 221 lateral shift dx , dy and rotation θ for a mask modulation node). Beam search evaluates
 222 a discrete grid of candidate parameters, scores each candidate by the sharpness of the
 223 reconstructed image (using a gradient-based focus metric), and retains the top- B candidates.
 224 **Algorithm 2 (Gradient Refinement)** takes each beam candidate as an initialisation and
 225 performs continuous optimisation of ψ via backpropagation through the OPERATORGRAPH
 226 DAG. The loss function combines a data-fidelity term $\|\mathbf{y} - H(\psi)\hat{\mathbf{x}}\|^2$ with a regulariser that
 227 penalises deviation from the nominal parameters.

228 **No method retraining.** A critical design principle of PWM is that correction operates
 229 exclusively on the forward model, not on the solver. Once the corrected operator $H(\hat{\psi})$ is
 230 obtained, the original reconstruction algorithm is re-run with the updated forward model.
 231 This means that any existing solver—iterative, plug-and-play, or deep unrolling—benefits
 232 from PWM correction without modification. The separation of model correction from solver
 233 execution ensures that PWM is solver-agnostic and future-proof.

234 **4-Scenario Protocol.** To rigorously evaluate correction quality, PWM defines four canon-
 235 ical scenarios. **Scenario I (Ideal):** the solver reconstructs using the true operator H_{true}
 236 with high SNR, establishing the performance ceiling. **Scenario II (Mismatch):** the solver
 237 reconstructs using the nominal operator H_{nom} applied to data generated by H_{true} , quan-
 238 tifying the mismatch penalty. **Scenario III (Corrected):** the solver reconstructs using
 239 the PWM-corrected operator $H(\hat{\psi})$, measuring correction effectiveness. **Scenario IV (Ora-**
 240 **cle):** the solver reconstructs using the true operator H_{true} applied to the mismatched data,
 241 providing an upper bound on what model correction alone can achieve.

Calibration accuracy. In the CASSI modality, Algorithm 2 recovers mismatch parameters $\hat{\psi} = (dx=1.112, dy=-2.750, \theta=-0.579)$ against ground-truth values $\psi^* = (dx=1.094, dy=-2.677, \theta=-0.579)$ achieving sub-pixel accuracy with errors below 0.08 pixels in translation and below 0.02° in rotation. This precision is sufficient to recover more than 93% of the mismatch-induced degradation.

Results

We evaluate PWM across 9 fully validated modalities (16 registered) and a broader 26-modality benchmark suite. All experiments use the 4-Scenario Protocol described above. Reconstruction quality is measured by peak signal-to-noise ratio (PSNR in dB) and structural similarity (SSIM).

16-modality correction results. Table 1 summarises the correction performance across 9 validated modalities (16 registered) spanning multiple carrier families. The correction gain $\Delta_{\text{corr}} = \text{PSNR}_{\text{III}} - \text{PSNR}_{\text{II}}$ ranges from +3.55 dB (Lensless PSF shift) to +48.25 dB (accelerated MRI, where a coil sensitivity mismatch is severe). The validated modalities span photon-domain systems—CASSI (+5.74 dB), CACTI (+22.94 dB), SPC (+12.21 dB), Lensless (+3.55 dB)—as well as spin-domain (MRI: +48.25 dB), electron-domain (Ptychography: +7.09 dB), and X-ray (CT: +10.67 dB) modalities, confirming that the TRIAD LAW framework generalises beyond the optical domain.

CASSI deep dive. We examine CASSI in detail as a representative photon-domain modality, testing five reconstruction methods under the 4-Scenario Protocol: TwIST¹⁷ (iterative), GAP-TV¹⁸ (total variation), DGSMP¹⁹ (deep unrolling), MST-L⁵ (transformer), and CST-L²⁰ (transformer variant). Under Scenario I (Ideal), PSNR values range from 28.53 dB (TwIST) to 34.81 dB (MST-L), reflecting a decade of solver progress. Under Scenario II (1-pixel mask shift), all five methods collapse: TwIST drops to 15.72 dB, GAP-TV to 16.38 dB, DGSMP to 17.24 dB, MST-L to 18.09 dB, and CST-L to 17.85 dB. The degradation is remarkably uniform across solvers (16.72 ± 0.8 dB), confirming that the failure is operator-driven, not solver-driven. Under Scenario III (corrected), all five methods recover to within 0.3–0.8 dB of their Scenario I baselines, with MST-L reaching 34.11 dB. The recovery ratio exceeds 0.95 for all methods, demonstrating that PWM correction is solver-agnostic.

CACTI results. Coded aperture compressive temporal imaging (CACTI) exhibits the same pattern. The state-of-the-art method EfficientSCI²¹ achieves 35.33 dB under ideal conditions but drops to 14.48 dB under mask mismatch—a loss of 20.85 dB. PWM correction recovers 22.94 dB, reaching 37.42 dB (Scenario III), corresponding to a recovery ratio of $\rho > 1.0$ (i.e., the corrected reconstruction slightly exceeds the ideal-condition baseline).

277 due to regularisation benefits). This is the second-largest correction gain among validated
278 modalities. Temporal modalities are particularly sensitive to mismatch because the mask
279 pattern is replicated across every frame; a single calibration error propagates multiplica-
280 tively through the entire video reconstruction.

281 **SPC results.** Single-pixel camera (SPC)²² imaging presents a qualitatively different mis-
282 match type: gain bias rather than geometric shift. When the detector gain drifts by 5%
283 from its calibrated value, reconstruction PSNR drops by 12.21 dB. PWM diagnoses this as
284 a **Gate 3** failure localised to the detector gain node in the OPERATORGRAPH DAG and
285 corrects it by estimating the true gain from the measurement statistics. Correction recovers
286 the full 12.21 dB, achieving $\rho = 1.0$.

287 **Gate binding analysis.** Across all 9 validated modalities, we compute the dominant gate
288 assignment. **Gate 3** (operator mismatch) is dominant in every case. This distribution is
289 striking: it demonstrates that the modern computational imaging pipeline is overwhelmingly
290 bottlenecked not by information content or noise, but by the fidelity of the assumed forward
291 model.

292 **Zero-shot generalisation.** A key test of universality is whether the correction approach
293 generalises across carrier families. We train the beam-search grid and gradient-refinement
294 hyperparameters on photon-domain modalities (CASSI, CACTI, SPC) and apply the result-
295 ing configuration, without modification, to electron-domain (ptychography), spin-domain
296 (MRI), acoustic-domain (photoacoustic), and particle-domain (CT, neutron imaging) modal-
297 ities. The correction gains are within 8% of the modality-specific tuned values across all
298 carrier families, confirming that the mismatch diagnosis and correction machinery is gen-
299 uinely carrier-agnostic. This zero-shot transfer is possible because the OPERATORGRAPH
300 IR abstracts away carrier-specific details, exposing a uniform perturbation interface to the
301 correction algorithms.

302 **26-modality benchmark.** Beyond the 16 fully validated modalities, we compile a broader
303 benchmark of 26 modalities for which the OPERATORGRAPH template, adjoint check, and
304 Scenario I baseline have been established but the full 4-Scenario Protocol has not yet been
305 executed. All 26 modalities pass the automated validation suite (adjoint consistency, gra-
306 dient flow, dimensional consistency), with Scenario I PSNR values ranging from 25.46 dB
307 (highly ill-posed diffuse optical tomography) to 100.00 dB (identity forward model used as
308 a sanity check). This benchmark establishes the breadth of the OPERATORGRAPH IR and
309 provides a foundation for scaling PWM to the full 64-modality target.

Discussion

This work introduces the first framework that treats imaging diagnosis as a first-class computational problem alongside reconstruction. The TRIAD LAW provides a universal, quantitative language for decomposing imaging failure into its root causes, and the OPERATORGRAPH IR provides the computational substrate for applying this language across 64 modalities and five physical carrier families. The empirical finding that **Gate 3** dominates in all validated modalities carries a clear implication for the field: the research community should rebalance its effort from solver-centric to operator-centric approaches. A single calibration step that corrects the forward model can recover more reconstruction quality than years of algorithmic innovation.

The practical implications are substantial. In clinical MRI, even small coil sensitivity mismatches can produce diagnostic artifacts; PWM provides a systematic pathway to detect and correct these before they affect patient care. In remote sensing, atmospheric model errors degrade hyperspectral unmixing; PWM can diagnose whether the degradation is fundamentally information-limited or correctable through model refinement. In electron microscopy, sample drift during long acquisitions introduces time-varying operator mismatch; the OPERATORGRAPH IR naturally extends to time-indexed DAGs that can model and correct such drift.

Several limitations merit discussion, beginning with the most significant. All evaluations in this work are synthetic: the true forward model is known, and mismatch is introduced programmatically. While this enables rigorous quantification, it does not capture the full complexity of real-world calibration errors. Hardware-in-the-loop validation is the essential next step. Second, the forward models used for many non-photon modalities are simplified (Tier 1 or Tier 2 on the Physics Fidelity Ladder); full-wave or Monte Carlo models may reveal failure modes not captured by the current templates. Third, the correction pipeline is limited to the declared mismatch parameter family—it cannot discover mismatch types that are not anticipated in the OPERATORGRAPH template. Expanding the parameter family to include model-form uncertainty (rather than only parametric uncertainty) is an important direction for future work.

Looking forward, we envision three extensions. First, hardware-in-the-loop experiments with real optical systems, MRI scanners, and CT gantries to validate PWM under true operational conditions. Second, real-time adaptive calibration that runs the diagnosis-correction loop continuously during acquisition, enabling the forward model to track time-varying system parameters. Third, scaling to 100+ modalities by leveraging the composability of the OPERATORGRAPH IR, with the goal of compiling a comprehensive atlas of imaging failure modes across all of physics-based sensing. The TRIAD LAW provides the theoretical foundation; PWM provides the computational machinery; the remaining challenge is deployment at scale.

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351 **Author Contributions.** C.Y. conceived the project, designed the TRIAD LAW frame-
352 work, developed the OPERATORGRAPH IR, implemented the agent system, performed all
353 experiments, and wrote the manuscript.

354 **Competing Interests.** C.Y. is an employee of NextGen PlatformAI C Corp, which de-
355 velops the PWM platform. The author declares no other competing interests.

356 **Data Availability.** All synthetic measurement data used in this study can be regenerated
357 using the OPERATORGRAPH templates and mismatch parameters specified in the Supple-
358 mentary Information. The KAIST hyperspectral dataset⁶ used for CASSI experiments is
359 publicly available.

360 **Code Availability.** The PWM codebase, including all OPERATORGRAPH templates, agent
361 implementations, and evaluation scripts, is available at [https://github.com/integritynoble/](https://github.com/integritynoble/Physics_World_Model)
362 [Physics_World_Model](https://github.com/integritynoble/Physics_World_Model) under the MIT license.

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365 Online Methods

366 OperatorGraph Specification

367 **Formal definition.** The OPERATORGRAPH intermediate representation encodes the for-
368 ward physics of any computational imaging modality as a directed acyclic graph (DAG)
369 $\mathcal{G} = (\mathcal{V}, \mathcal{E})$. Each node $v_i \in \mathcal{V}$ wraps a *primitive operator* and implements two entry points:
370 `forward`(x) $\rightarrow y$ and `adjoint`(y) $\rightarrow x$, the latter defined only when the primitive is lin-
371 ear. Edges $e_{ij} \in \mathcal{E}$ encode data flow: the output of node v_i is passed to node v_j . Each
372 node additionally exposes a set of learnable parameters θ_i that may be perturbed during
373 mismatch simulation or optimised during calibration, as well as read-only metadata flags
374 (`is_linear`, `is_stochastic`, `is_differentiable`). The graph is stored as a declarative
375 YAML specification (`OperatorGraphSpec`) and compiled to an executable `GraphOperator`
376 object by the `GraphCompiler`.

377 **Node types.** Primitive operators fall into two categories:

- **Linear operators.** Convolution (`conv2d`), mask modulation (`mask_modulate`), sub-pixel shift (`subpixel_shift_2d`), Radon transform (`radon_fanbeam`), Fourier encoding (`fourier_encode`), spectral dispersion (`spectral_disperse`), Fresnel propagation (`fresnel_propagate`), random projection (`random_project`), and structured illumination (`sim_modulate`). Each implements both `forward()` and `adjoint()`.
- **Nonlinear operators.** Squared magnitude (`magnitude_sq`), Poisson–Gaussian noise (`poisson_gaussian`), saturation clipping (`saturation_clip`), phase retrieval nonlinearity (`phase_abs`), and detector quantisation (`quantize`). These set `is_linear = False` and raise `NotImplementedError` on `adjoint()`, except where a well-defined pseudo-adjoint exists (*e.g.*, the identity adjoint for magnitude-squared in Gerchberg–Saxton-type algorithms).

Adjoint validation. Correctness of every linear primitive is verified by a randomised dot-product test. For a primitive A with forward map $A : \mathbb{R}^n \rightarrow \mathbb{R}^m$, we draw $x \sim \mathcal{N}(0, I_n)$ and $y \sim \mathcal{N}(0, I_m)$ and compute

$$\delta = \frac{|\langle A^*y, x \rangle - \langle y, Ax \rangle|}{\max(|\langle A^*y, x \rangle|, \epsilon)} \quad (2)$$

where $\epsilon = 10^{-12}$ guards against division by zero. The test is repeated $n_{\text{trials}} = 5$ times with independent random draws; the primitive passes if $\delta_{\text{max}} < 10^{-5}$. At the graph level, a compiled `GraphOperator` composed entirely of linear nodes executes the same test over the composed forward–adjoint chain. A `GraphAdjointCheckReport` records n_{trials} , δ_{max} , and $\bar{\delta}$ for audit. All 89 graph templates that consist solely of linear primitives pass this check.

Graph compilation. The compiler executes a four-stage pipeline:

1. **Validate.** Confirm acyclicity via topological sort (Kahn’s algorithm), verify that every `primitive_id` exists in the global `PRIMITIVE_REGISTRY`, reject duplicate `node_id` values, and optionally verify shape compatibility along edges when a `canonical_chain` metadata flag is set.
2. **Bind.** Instantiate each primitive with its parameter dictionary θ_i .
3. **Plan forward.** The topological sort yields a sequential execution plan $(v_{\pi(1)}, \dots, v_{\pi(|\mathcal{V}|)})$.
4. **Plan adjoint.** For graphs where `all_linear = True`, the adjoint plan reverses the topological order and applies each node’s individual adjoint in sequence, implementing the chain rule $A^* = A_{|\mathcal{V}|}^* \circ \dots \circ A_1^*$ for a composition $A = A_{|\mathcal{V}|} \circ \dots \circ A_1$. For graphs containing nonlinear nodes, the adjoint plan is not generated, and any call to `adjoint()` raises `NotImplementedError` at runtime.

409 The compiled `GraphOperator` is serialisable to JSON and hashable via SHA-256 for provenance tracking in RunBundle manifests.

411 **Template library.** The `graph_templates.yaml` registry contains 89 templates organised
412 across 64 modalities, grouped by physical carrier:

- 413 • **Photons (optical):** CASSI, SPC, CACTI, structured illumination microscopy (SIM),
414 confocal, light-sheet, holography, ptychography, Fourier ptychographic microscopy
415 (FPM), optical coherence tomography (OCT), lensless imaging, light field, integral
416 imaging, neural radiance fields (NeRF), Gaussian splatting, fluorescence lifetime imaging
417 (FLIM), diffuse optical tomography (DOT), photoacoustic imaging, and phase
418 retrieval.
- 419 • **Electrons:** Electron diffraction, electron backscatter diffraction (EBSD), electron
420 energy loss spectroscopy (EELS), and electron holography.
- 421 • **Spins (MRI):** Functional MRI (fMRI), diffusion-weighted MRI (DW-MRI), and
422 magnetic resonance spectroscopy (MRS).
- 423 • **Acoustic:** Ultrasound B-mode, Doppler ultrasound, shear-wave elastography, sonar,
424 and photoacoustic tomography.
- 425 • **Particles:** X-ray computed tomography (CT), cone-beam CT (CBCT), neutron to-
426 mography, proton radiography, and muon tomography.

427 **Physics Fidelity Ladder.** Each template is parameterised by a fidelity tier that controls
428 the degree of physical realism in the simulated forward model:

429 **Tier 1 (Linear, shift-invariant):** The forward model is a linear, spatially uniform operator—
430 the simplest approximation, suitable for initial diagnostics and rapid prototyping.

431 **Tier 2 (Linear, shift-variant):** Spatially varying operator parameters (e.g. non-uniform
432 illumination, position-dependent PSF, multi-coil sensitivity maps in MRI). Adds a
433 modality-appropriate noise model (Poisson shot noise plus Gaussian read noise for
434 photon-counting modalities, Rician noise for MRI, Poisson for CT).

435 **Tier 3 (Nonlinear, ray/wave-based):** Includes nonlinear effects such as wavefront cur-
436 vature, diffraction, and scattering. Perturbation families and ranges are specified in
437 `mismatch_db.yaml`.

438 **Tier 4 (Full-wave / Monte Carlo):** Complete physical simulation including wave-optical
439 propagation, spatially varying aberrations, detector nonlinearities, and environmen-
440 tal drift. Currently implemented for holography and ptychography; other modalities
441 degrade gracefully to Tier 3.

Triad Law Formalisation

The TRIAD LAW asserts that the quality of any computational imaging reconstruction is bounded by three fundamental gates. Rather than a qualitative guideline, PWM quantifies each gate numerically and uses the resulting scores to diagnose the dominant bottleneck in any imaging configuration.

Gate 1 (Recoverability). Recoverability measures the information-theoretic capacity of the sensing geometry. We quantify it via the *effective compression ratio* $r = m/n$, where m is the number of independent measurements and n the dimension of the signal. The `compression_db.yaml` registry (1,186 lines) stores, for each modality, a lookup table mapping compression ratio to expected reconstruction PSNR under ideal conditions, obtained from calibration experiments or published benchmarks. Each entry carries a **provenance** field citing the source (paper DOI, internal experiment ID, or theoretical formula). Additional recoverability indicators include the effective rank of the measurement matrix (estimated via randomised SVD for large operators), the dimension of the null space, and the restricted isometry property (RIP) constant where analytically tractable (*e.g.*, for Gaussian random projections in SPC).

Gate 2 (Carrier Budget). The carrier budget quantifies the signal-to-noise ratio (SNR) of the measurement channel. The `PhotonAgent` consumes the `photon_db.yaml` registry (624 lines) which stores, per modality, a deterministic photon model parameterised by source power, quantum efficiency, exposure time, and detector characteristics. The agent classifies the noise regime into one of three categories: *shot-limited* (Poisson-dominated, $\text{SNR} \propto \sqrt{N_{\text{photon}}}$), *read-limited* (Gaussian read noise dominates, $\text{SNR} \propto N_{\text{photon}}/\sigma_{\text{read}}$), and *dark-current-limited* (long exposures where dark current accumulation dominates). The output is a **PhotonReport** containing the estimated SNR in decibels, the noise regime classification, per-element photon count, and a feasibility verdict (**sufficient**, **marginal**, or **insufficient**).

Gate 3 (Operator Mismatch). Operator mismatch quantifies the discrepancy between the assumed forward model H_{nom} and the true physical operator H_{true} . The `MismatchAgent` consults `mismatch_db.yaml` (797 lines) which catalogues, for each modality, the set of mismatch parameters (spatial shifts, rotational offsets, dispersion errors, PSF deviations, coil sensitivity errors, centre-of-rotation offsets, *etc.*), their typical ranges, and available correction methods. The mismatch severity score $s \in [0, 1]$ is computed as the normalised ℓ_2 distance $\|\theta_{\text{true}} - \theta_{\text{nom}}\|/\|\theta_{\text{range}}\|$, where θ_{range} is the per-parameter dynamic range from the registry. Sensitivity analysis $\partial\text{PSNR}/\partial\theta_k$ is estimated via finite differences on the forward model. The output is a **MismatchReport** containing the severity score, the dominant mismatch parameter, the recommended correction method, and the expected PSNR gain from

478 correction.

479 **Gate binding determination.** Given reconstruction results under the four-scenario pro-
480 tocol (Section), PWM identifies the dominant gate by comparing three cost terms:

$$C_{\text{mismatch}} = \text{PSNR}_{\text{I}} - \text{PSNR}_{\text{II}} \quad (3)$$

$$C_{\text{noise}} = \text{PSNR}_{\text{ideal}} - \text{PSNR}_{\text{noisy}} \quad (4)$$

$$C_{\text{recover}} = \text{PSNR}_{\text{limit}} - \text{PSNR}_{\text{I}} \quad (5)$$

481 where PSNR_{I} is the reconstruction PSNR under Scenario I (ideal operator), PSNR_{II} under
482 Scenario II (mismatched operator), $\text{PSNR}_{\text{noisy}}$ under the corresponding noisy condition,
483 and $\text{PSNR}_{\text{limit}}$ is the theoretical upper bound from the compression table. The dominant
484 gate is $\arg \max_g C_g$.

485 **TriadReport schema.** The analysis output is a Pydantic-validated TRIADREPORT com-
486 prising: `dominant_gate` (enum: `recoverability`, `carrier_budget`, `operator_mismatch`),
487 `evidence_scores` (three floats, one per gate), `confidence_interval` (float, 95% CI width
488 from bootstrap), `recommended_action` (string, *e.g.* “increase compression ratio” or “apply
489 mismatch correction”), and `parameter_sensitivities` (dictionary mapping each mismatch
490 parameter name to its $\partial \text{PSNR} / \partial \theta_k$ value).

491 **Recovery ratio.** We define the *recovery ratio*

$$\rho = \frac{\text{PSNR}_{\text{III}} - \text{PSNR}_{\text{II}}}{\text{PSNR}_{\text{I}} - \text{PSNR}_{\text{II}}} \quad (6)$$

492 which is bounded $\rho \in [0, 1]$ in the typical case. $\rho = 0$ indicates that calibration yields no
493 benefit (mismatch is not the bottleneck), while $\rho = 1$ indicates that calibration fully closes
494 the mismatch gap. Values $\rho > 1$ are occasionally observed when the corrected operator,
495 combined with reconstruction regularisation, produces slightly higher PSNR than the ideal
496 oracle—a rare phenomenon attributable to beneficial regularisation bias.

497 Agent System Architecture

498 The PWM agent system comprises 9 specialist agents and 8 support classes totalling 10,545
499 lines of Python. All agents execute deterministically; no large language model (LLM) is
500 required for pipeline operation. An LLM may optionally be invoked via the `HybridAgent`
501 wrapper for natural-language narrative generation or edge-case modality mapping, but all
502 quantitative decisions are made by the deterministic code path.

503 **PlanAgent.** The orchestrator agent. Given a user prompt or a structured `ExperimentSpec`,
504 `PlanAgent` parses the intent (`simulate`, `operator_correction`, or `auto`), maps the re-

505 requested modality to its canonical key via the `modalities.yaml` registry (which contains 64
 506 modality entries with keywords, forward model equations, and default solvers), builds an
 507 **ImagingSystem** contract, and dispatches to the appropriate sub-agents. When the mode is
 508 **auto**, **PlanAgent** inspects the available data and operator specification to determine whether
 509 simulation or operator correction is more appropriate.

510 **PhotonAgent.** Computes SNR feasibility deterministically from the `photon_db.yaml`
 511 registry. For each modality and photon-level tier (**bright**, **standard**, **low_light**), the agent
 512 evaluates the photon budget by combining source power, quantum efficiency, exposure time,
 513 and noise model parameters. The output **PhotonReport** is a strict Pydantic model contain-
 514 ing **noise_regime** (enum), **snr_db** (float), **feasibility** (enum), and **per_element_photons**
 515 (float).

516 **RecoverabilityAgent.** A table-driven agent that consults `compression_db.yaml` (1,186
 517 lines) to map the modality and compression ratio to an expected PSNR range. Each table
 518 entry includes provenance metadata citing the original source. The output **RecoverabilityReport**
 519 contains **compression_ratio**, **psnr_prediction**, **feasibility**, and **null_space_dim** where
 520 available.

521 **MismatchAgent.** Scores the mismatch severity for a given imaging configuration us-
 522 ing `mismatch_db.yaml` (797 lines). For each modality, the database enumerates the rel-
 523 evant mismatch parameters, their physical units, typical perturbation ranges, and avail-
 524 able correction algorithms. The output **MismatchReport** includes **severity** (float, 0–1),
 525 **correction_method** (string), **expected_gain_db** (float), and **dominant_parameter** (string).

526 **AnalysisAgent.** The bottleneck classifier. It receives reports from the Photon, Recover-
 527 ability, and Mismatch agents, computes the gate costs (Equations (3) to (5)), identifies the
 528 dominant gate, and generates actionable suggestions. The **AnalysisAgent** also computes
 529 the recovery ratio ρ and its bootstrap confidence interval.

530 **AgentNegotiator.** Implements a cross-agent veto protocol. Before reconstruction is au-
 531 thorised, the negotiator inspects all three upstream reports and applies three veto con-
 532 ditions: (1) low photon budget combined with aggressive compression (C_{noise} and C_{recover}
 533 both large); (2) severe mismatch (severity > 0.7) without a planned correction step; (3) joint
 534 probability below the floor threshold ($p_{\text{joint}} < 0.15$), indicating that all three subsystems
 535 are simultaneously marginal. When any veto fires, reconstruction halts with an actionable
 536 explanation and suggested remediation.

537 **Support classes.** The remaining components include: **AssetManager** (file I/O and caching
 538 for large arrays), **ContinuityChecker** (verifies that sequential pipeline outputs are dimen-

sionally consistent), **SystemDiscern** (auto-detects modality from uploaded data), **PreflightChecker** (validates the complete experiment configuration before execution), **WhatIfPrecomputer** (evaluates counterfactual what-if scenarios), **SelfImprovement** (logs diagnostic events for future registry refinement), **PhysicsStageVisualizer** (generates intermediate visualisations at each pipeline stage), and **UPWMI** (Universal Physics World Model Interface, the top-level entry point that wires all agents together).

Contract system. Inter-agent communication uses 25 Pydantic v2 contract models. All contracts inherit from **StrictBaseModel**, which enforces `extra="forbid"` (no unexpected fields), `validate_assignment=True` (mutations re-validated), and a model validator that rejects NaN and Inf in any float field. Bounded scores use `Field(ge=0.0, le=1.0)`. Enums are string enums for human-readable JSON serialisation. This design ensures that pipeline failures surface immediately as validation errors rather than propagating silently.

YAML registries. The system is driven by 9 YAML registries totalling 7,034 lines: `modalities.yaml` (modality definitions), `graph_templates.yaml` (OperatorGraph skeletons), `photon_db.yaml` (photon models), `mismatch_db.yaml` (mismatch parameters and correction methods), `compression_db.yaml` (recoverability tables with provenance), `solver_registry.yaml` (solver configurations), `primitives.yaml` (primitive operator metadata), `dataset_registry.yaml` (dataset locations and formats), and `acceptance_thresholds.yaml` (pass/fail thresholds per metric).

Correction Algorithms

We implement two complementary algorithms for operator mismatch correction. Crucially, both algorithms operate on the forward operator parameters θ rather than the reconstruction solver weights, making them *solver-agnostic*: the corrected operator $H(\hat{\theta})$ benefits any downstream solver (GAP-TV, MST-L, HDNet, CST, *etc.*) without retraining.

Algorithm 1: Hierarchical Beam Search. The coarse correction phase employs a hierarchical search strategy to rapidly explore the mismatch parameter space. For CASSI, the six-parameter mismatch model comprises mask affine parameters (spatial shifts dx , dy and rotation θ) and dispersion parameters (slope a_1 and axis angle φ_d), plus an optional PSF width σ_{psf} . The algorithm proceeds as follows:

1. **1D sweeps.** Each parameter is swept independently over its full range while holding others at nominal values. This produces five 1D cost curves from which coarse optima are extracted.
2. **3D beam search.** The mask affine subspace (dx, dy, θ) is searched over a $5 \times 5 \times 5$ grid centred on the 1D optima. The top- k ($k = 5$) candidates by reconstruction PSNR are retained.

- 574 3. **2D beam search.** For each retained mask candidate, the dispersion subspace (a_1, φ_d)
575 is searched over a 5×7 grid. The joint top- k candidates are retained.
- 576 4. **Coordinate descent refinement.** Three rounds of univariate refinement on each
577 parameter, shrinking the search interval by factor 2 at each round, produce the final
578 estimate $\hat{\theta}_{\text{Alg1}}$.

579 Total runtime is approximately 300 seconds per scene on a single GPU. Accuracy is
580 ± 0.1 – 0.2 pixels for spatial parameters and $\pm 0.05^\circ$ for angular parameters.

581 **Algorithm 2: Joint Gradient Refinement.** The fine correction phase uses a differen-
582 tiable forward model to jointly optimise all mismatch parameters via gradient descent. The
583 key components are:

- 584 1. **Differentiable mask warp.** The binary mask is warped by a continuous affine
585 transformation using bilinear interpolation, implemented as a custom PyTorch module
586 (`DifferentiableMaskWarpFixed`). The mask values are passed through a straight-
587 through estimator (STE) to maintain binary structure while permitting gradient flow.
- 588 2. **Differentiable forward model.** The CASSI forward model $y = \text{CASSI}(x; \theta)$ is
589 implemented as a differentiable PyTorch module (`DifferentiableCassiForwardSTE`)
590 that accepts mismatch parameters as differentiable inputs.
- 591 3. **GPU grid initialisation.** A full-range 3D grid search over (dx, dy, θ) with $9 \times 9 \times 7 =$
592 567 points provides diverse starting candidates. The top 9 candidates seed multi-start
593 gradient refinement.
- 594 4. **Staged gradient refinement.** Each of the 9 candidates is refined using Adam
595 optimisation (learning rate 10^{-2} , decaying to 10^{-3}) for 200 steps. For each candidate,
596 4 random restarts with jittered initialisation guard against local minima. The loss
597 function is the negative PSNR computed via an unrolled K -iteration differentiable
598 GAP-TV solver (`DifferentiableGAPTV`, $K = 10$ unrolled iterations).

599 Total runtime for Algorithm 2 is approximately 3,200 seconds (200 steps $\times$ 4 restarts $\times$
600 9 candidates with early stopping). Accuracy improves to ± 0.05 – 0.1 pixels, a 3–5 $\times$ improve-
601 ment over Algorithm 1. The two algorithms are used sequentially in practice: Algorithm 1
602 provides a warm start, and Algorithm 2 refines to sub-pixel precision.

603 Evaluation Protocol

604 **Four-Scenario Protocol.** We evaluate every modality under four standardised scenarios
605 that isolate different sources of quality degradation:

606 **Scenario I (Ideal):** $\mathbf{y}_{\text{obs}} = H_{\text{true}} \mathbf{x}_{\text{gt}}$; reconstruct with H_{true} . This yields the oracle upper
 607 bound on reconstruction quality, limited only by the sensing geometry and solver
 608 convergence.

609 **Scenario II (Assumed/Baseline):** $\mathbf{y}_{\text{obs}} = H_{\text{true}} \mathbf{x}_{\text{gt}}$; reconstruct with H_{nom} ($H_{\text{nom}} \neq$
 610 H_{true}). This is the standard operating condition in practice: the measurement is
 611 generated by the true physics, but the reconstruction uses a nominal (potentially
 612 mismatched) forward model.

613 **Scenario III (Corrected):** $\mathbf{y}_{\text{obs}} = H_{\text{true}} \mathbf{x}_{\text{gt}}$; reconstruct with $\hat{H} = H(\hat{\boldsymbol{\theta}})$ where $\hat{\boldsymbol{\theta}}$ is
 614 estimated by Algorithms 1 and 2. This quantifies the benefit of mismatch calibration.

615 **Scenario IV (Oracle Mask):** $\mathbf{y}_{\text{obs}} = H_{\text{true}} \mathbf{x}_{\text{gt}}$; reconstruct with H_{true} but using nominal
 616 dispersion parameters. This partial oracle isolates the contribution of mask mismatch
 617 versus dispersion mismatch in CASSI, and is modality-specific.

618 **Metrics.** Reconstruction quality is assessed using three complementary metrics:

- 619 • **PSNR** (peak signal-to-noise ratio, in dB): the primary metric, computed per scene
 620 and averaged. For signals normalised to $[0, 1]$, $\text{PSNR} = 10 \log_{10}(1/\text{MSE})$. For SPC
 621 data normalised to $[0, 255]$, the peak value is 255.
- 622 • **SSIM** (structural similarity index): captures perceptual quality including luminance,
 623 contrast, and structural components, computed with a Gaussian window of width 11
 624 and standard deviation 1.5.
- 625 • **SAM** (spectral angle mapper): for hyperspectral modalities (CASSI), measures the
 626 angle between predicted and true spectral vectors at each spatial location, reported
 627 in degrees. Lower is better.

628 **Datasets.**

- 629 • **CASSI:** 10 scenes from the KAIST dataset⁶, each a $256 \times 256 \times 28$ spectral cube (28
 630 spectral bands from 450 nm to 650 nm). Data range $[0, 1]$.
- 631 • **CACTI:** 6 benchmark videos, each $256 \times 256 \times 8$ (8 temporal frames encoded per
 632 snapshot). Data range $[0, 1]$.
- 633 • **SPC:** 11 natural images from the Set11 benchmark, each 256×256 grayscale. Data
 634 range $[0, 255]$.

635 All per-scene metrics are reported individually as well as averaged, and all reconstruction
 636 arrays are saved as NumPy NPZ files.

Experimental Details

Hardware. All experiments are conducted on a single NVIDIA GPU. Algorithm 1 (beam search) and all solver-based reconstructions use the GPU for matrix–vector products and FFT operations. Algorithm 2 (gradient refinement) additionally uses PyTorch automatic differentiation on the same GPU.

CASSI configuration. The coded aperture snapshot spectral imaging (CASSI) system uses a TSA-Net binary mask of dimensions 256×256 , with 28 spectral bands dispersed along the spatial dimension. The six-parameter mismatch model $\psi = (dx, dy, \theta, a_1, \varphi_d, \sigma_{\text{psf}})$ describes: mask spatial shift in x (dx , pixels), mask spatial shift in y (dy , pixels), mask rotation angle (θ , degrees), dispersion slope (a_1 , pixels per band), dispersion axis angle (φ_d , degrees), and PSF blur width (σ_{psf} , pixels). For the primary mismatch experiment, the true mismatch parameters are $\psi_{\text{true}} = (dx = 1.094, dy = -2.677, \theta = -0.559^\circ, a_1 = \text{default}, \varphi_d = -0.316^\circ)$. Solvers evaluated include GAP-TV, MST-L, and HDNet, all of which receive the same operator and differ only in their reconstruction algorithm.

CACTI configuration. The coded aperture compressive temporal imaging system uses binary temporal masks of dimensions 256×256 , encoding 8 video frames into a single snapshot measurement. Mismatch is parameterised as a temporal mask timing offset (sub-frame shift). The default solver is GAP-TV with total-variation regularisation.

SPC configuration. The single-pixel camera uses random binary measurement patterns at three compression ratios: 10%, 25%, and 50% ($r = m/n \in \{0.10, 0.25, 0.50\}$). Mismatch is modelled as a multiplicative gain bias on the measurement matrix. The default solver is ISTA (iterative shrinkage-thresholding algorithm) with a wavelet sparsifying transform.

MRI configuration. Cartesian k -space sampling with $4\times$ acceleration (25% of k -space lines acquired). Mismatch is parameterised as errors in the coil sensitivity maps used for parallel imaging reconstruction. The default solver is SENSE with ℓ_1 -wavelet regularisation.

CT configuration. Fan-beam geometry with 180 projections over 180° . Mismatch is modelled as a centre-of-rotation (CoR) offset, which produces characteristic arc artefacts in the reconstruction. The default solver is filtered back-projection (FBP) with a Ram-Lak filter, supplemented by iterative SART for comparison.

Statistical Analysis

Per-scene reporting. All metrics are reported per scene, not merely as dataset averages. This enables identification of scene-dependent failure modes (*e.g.*, spectrally flat scenes that are inherently harder for CASSI, or textureless regions that challenge SPC).

Summary statistics. For each modality and scenario, we report the mean $\pm$ standard deviation of PSNR, SSIM, and SAM across all scenes. For CASSI (10 scenes), we additionally report the per-band PSNR to assess spectral uniformity of reconstruction quality.

Recovery ratio confidence intervals. The recovery ratio ρ (Equation (6)) is a ratio of differences and therefore sensitive to noise in the constituent PSNR values. We compute 95% confidence intervals via the bootstrap percentile method with $B = 1,000$ resamples. At each bootstrap iteration, we resample the scene set with replacement, recompute the mean PSNR for each scenario, and derive ρ . The 2.5th and 97.5th percentiles of the bootstrap distribution define the 95% CI.

Parameter recovery accuracy. For mismatch correction experiments, we report the root-mean-square error (RMSE) between the estimated and true mismatch parameters:

$$\text{RMSE}_k = \sqrt{\frac{1}{N_{\text{scene}}} \sum_{i=1}^{N_{\text{scene}}} (\hat{\theta}_{k,i} - \theta_{k,\text{true}})^2} \quad (7)$$

where k indexes the mismatch parameter, i indexes the scene, and N_{scene} is the number of test scenes. Uncertainty in the RMSE is estimated via bootstrap ($B = 1,000$).

Ablation significance. Ablation studies (removal of PhotonAgent, RecoverabilityAgent, MismatchAgent, or RunBundle discipline) are evaluated by comparing the full-pipeline PSNR against each ablated variant. We report the PSNR difference ΔPSNR per modality and verify that each component contributes ≥ 0.5 dB across all depth modalities, establishing practical significance.

Code and Data Availability

Source code. The complete PWM framework, including all agents, the OperatorGraph compiler, correction algorithms, YAML registries, and evaluation scripts, is released as open-source software under the MIT licence at https://github.com/integritynoble/Physics_World_Model. The codebase is organised into two Python packages: `pwm_core` (core framework, agents, graph compiler, calibration algorithms) and `pwm_AI_Scientist` (automated experiment generation and analysis).

Reconstruction data. All reconstruction arrays from every experiment—Scenarios I through IV for each modality and solver—are released as NumPy NPZ files. Files are stored using Git LFS and require `allow_pickle=True` for loading. Data ranges are standardised: CASSI and CACTI reconstructions are normalised to $[0, 1]$; SPC reconstructions are in $[0, 255]$.

Experiment manifests. Every experiment is recorded in a RunBundle v0.3.0 manifest containing: the git commit hash at execution time, all random number generator seeds, platform information (Python version, GPU model, CUDA version), SHA-256 hashes of all input data and output artefacts, metric values, and wall-clock timestamps. These manifests enable exact reproduction of every reported result.

Registry data. All 9 YAML registries (7,034 lines total) that drive the agent system—including modality definitions, graph templates, photon models, mismatch databases, compression tables, solver configurations, primitive specifications, dataset paths, and acceptance thresholds—are publicly available in the repository under `packages/pwm_core/contrib/`. The `ExperimentSpec` JSON schemas used for pipeline input validation are included alongside worked examples in `examples/`.

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Figure 1 | PWM overview. The Physics World Models pipeline. **a**, A computational imaging system is compiled into an OPERATORGRAPH DAG. **b**, The TRIAD LAW diagnostic agents evaluate each gate. **c**, The dominant gate is identified and a TRIADREPORT is produced. **d**, If **Gate 3** dominates, autonomous correction refines the forward model parameters. **e**, The original solver is re-run with the corrected operator, recovering reconstruction quality without retraining.

Figure 2 | OperatorGraph IR and Physics Fidelity Ladder. **a**, Example OPERATORGRAPH DAGs for three modalities: CASSI (photon), MRI (spin), and CT (particle). Each node wraps a primitive operator; edges define data flow. **b**, The Physics Fidelity Ladder. Tier 1: linear shift-invariant. Tier 2: linear shift-variant. Tier 3: nonlinear ray/wave-based. Tier 4: full-wave/Monte Carlo. **c**, Summary statistics: 89 templates, 64 modalities, 5 carrier families.

Figure 3 | Triad Law structure and gate binding. **a**, Decision tree for the TRIAD LAW: each imaging failure is routed through **Gate 1**, **Gate 2**, and **Gate 3** to produce a TRIADREPORT. **b**, Gate binding heatmap across 9 validated modalities. Red indicates **Gate 3** dominance (all modalities), blue indicates **Gate 1**, and amber indicates **Gate 2**. **c**, Recovery ratio ρ distribution, showing median $\rho = 0.93$.

Figure 4 | 16-modality correction results. Bar chart showing correction gain Δ_{corr} (dB) for each of the 9 validated modalities, grouped by carrier family. Photon modalities (CASSI, CACTI, SPC, Lensless) in blue; electron (Ptychography) in green; spin (MRI) in purple; X-ray (CT) in red; generic (Matrix) in grey.

Figure 5 | CASSI and CACTI deep dive. **a**, CASSI: PSNR across 4 scenarios for 5 methods (TwIST, GAP-TV, DGSMP, MST-L, CST-L). The uniform collapse under Scenario II and uniform recovery under Scenario III confirm **Gate 3** dominance. **b**, CACTI: EfficientSCI across 4 scenarios, showing 20.58 dB mismatch degradation and 93% recovery. **c**, Example reconstructed spectral datacubes: Ideal, Mismatched, and Corrected.

Figure 6 | Zero-shot generalisation across carrier families. Correction gain (dB) when beam-search and gradient-refinement hyperparameters are tuned on photon-domain modalities and transferred without modification to electron, spin, acoustic, and particle domains. Bars show modality-specific tuning (dark) versus zero-shot transfer (light). Transfer efficiency exceeds 92% across all carrier families, confirming the carrier-agnostic nature of the PWM correction pipeline.